Math 199, Fall 2022
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Participation assignment 6 - Setting up definite integrals

Estimated time: Less than 1 hour.

Point value: 3 points.

Goals: The main goals of today's class will be to understand how to set up a definite integral to compute quantities you are interested in, and to practice recognizing when some methods are better than others.

Here's a reminder of our

General strategy for setting up a definite integral: To express a quantity, Q, as a definite integral, make a "local estimate" of the amount of Q corresponding to a very small interval $[v, v+dv]$, in a *conveniently chosen* variable, v. If this local estimate has the form $f(v)dv$, then

$$Q = \int_a^b f(v)dv,$$

where a and b are the minimum and maximum values of v that correspond to Q.

1) Find the area of the region above the graph of $y = |x|$ and below the graph of $y = \sqrt{1-x^2}$. Do this in *six different ways*. First, draw a picture of the region whose area you are trying to find.

(a) Recognize the region as a simple fraction of a shape that you know, and compute the area without calculus.

(b) Set up an integral that computes the area of the region, where you use x as your "chosen variable" (i.e. the variable of integration).

(c) Do the same as (b), but with y as the variable of integration.
Hint: you may have to write it as a sum of two integrals.

(d) Look at the graph of $y = |x|$ closely. If you rotate your drawing a little bit (or your head), you might recognize that this graph looks like a pair of axes. Label these new axes $\tilde{x}$ and $\tilde{y}$. Write down a new characterization of our region of interest using these coordinates. Repeat (b)/(c), but using either the variable $\tilde{x}$ or $\tilde{y}$.

(e) Consider the variable θ, which measures the angle traced out in the counterclockwise direction, starting from the positive x-axis. Repeat (b)/(c)/(d) using θ as the "conveniently chosen variable".

(f) Consider the variable r, which measures distance from the origin. Repeat using r as the "conveniently chosen variable".

2) Rank the different methods used in problem (1) from best to worst (in your opinion).

3) Based on your experience working through problem (1) and otherwise, what are the things to look for when trying to find a good and effective (and hopefully easy) way to set up a definite integral. How can you gauge the "convenience" of a certain variable versus other potential choices? Discuss with your group, and write down a guide in your own words, either on the back of this page, or in your notebook to reference later.